

Stock-Market Dynamics, Rev. 4: Four-Axis Regimes, Capital Weighting, and a Paper-Trading Protocol

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Objectives

- ▶ Four regime axes: momentum $\times$ volatility $\times$ turnover $\times$ size = 24 bins
- ▶ **Capital-weighted** bin shares (not headcounts)
- ▶ **True historical** S&P 500 membership — never backfilled
- ▶ Zero-lookahead walk-forward forecast $\rightarrow$ monthly **tilt portfolio** with frozen weights
- ▶ Benchmarks: SPY buy-and-hold; **equal-weight historical members**
- ▶ Paper record = **prerequisite** for any real-capital deployment

Data

- ▶ **Membership:** fja05680/sp500 public reconstruction, 2,720 snapshots 1996–2026 (*community, not official S&P*)
- ▶ **Prices:** Yahoo Finance, 659 tickers; aliases hand-validated (FB→META, UTX/RTN→RTX, ...)
- ▶ **Shares:** SEC EDGAR XBRL, 47.9k facts; starts mid-2009 — pre-2009 backfilled, *estimated*
- ▶ **Turnover** = dollar volume / mcap: a *measured liquidity proxy*, not true turnover
- ▶ Panel: 138,255 member/months; 80.7% binnable on all four axes

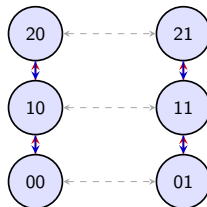
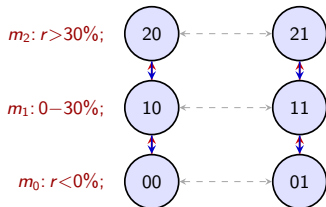
Four-axis reaction-admissibility graph

v_0 low vol

v_1 high vol

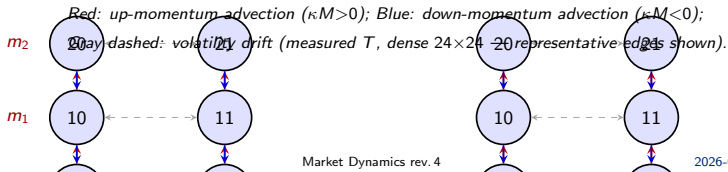
turnover LOW / size SMALL

turnover LOW / size LARGE



turnover HIGH / size SMALL

turnover HIGH / size LARGE



Master equation

$$\frac{d\mathbf{c}}{dt} = (T_{\text{eff}} - I)\mathbf{c} + A(M)\mathbf{c} + H(\mathbf{c})\mathbf{c} + \mathbf{e}$$

- ▶ **Drift:** cap-weighted T , calm vs. stress regimes — *measured*
- ▶ **Advection:** $M(t) = \bar{r}(t) - \bar{r}(t-12)$ translates momentum lanes; $\kappa = 1$ *mechanical*; reflecting extremes pile mass (spikes)
- ▶ **Herding / sticky drift:** calibrated $h_1 = \lambda_1 = 0$ on 2000–2007 — *off* in rev-4
- ▶ **Damper:** impedes extreme-bin outflow post-spike; $p = 1.2$, 6-mo memory, gate 0.5 — *assumed* (rev-3 episode tuning)

Driver forecast: mechanical nowcast

$$\hat{M}(t+1) = R_{\text{SPY}}(t) - R_{\text{SPY}}(t-12)$$

- ▶ From the momentum definition:
 $\text{mom}_i(t+1) - \text{mom}_i(t) \approx r_i(t) - r_i(t-12)$
- ▶ Both terms known at month-end t — **zero lookahead**
- ▶ 2008–2022: corr. 0.50, 70% directional, RMSE 0.074
- ▶ Expanding AR(1): corr. -0.02 , 41% directional — **discarded**

Conditional hindcasts

2008–09 (RMSE 0.1685)

- ▶ Bottom-lane peak: 0.88 pred. vs. 1.00 real
- ▶ Amplitude close, **3 months early**
- ▶ 2009 V-recovery under-predicted (uniform-bin limit)

2020–21 (RMSE 0.1110)

- ▶ Top-lane peak: 0.95 pred. vs. 0.92 real
- ▶ Within 0.03, one month late
- ▶ Realized volatile around peak; model smoother

Mechanism: M translates the distribution; reflecting extremes accumulate it. The model spikes when M spikes.

Paper-trading protocol

At each month-end t (2008–2022):

1. Compute $\hat{M}(t+1)$ mechanically
2. Simulate one step to $\hat{\mathbf{c}}(t+1)$
3. Tilt: $\pi_b \propto \max(c_b + \gamma(\hat{c}_b - c_b), 0)$, $\gamma = 1$ (*additive*)
4. Cap-weight stocks within bins; **freeze** through $t+1$
5. 5 bps one-way on turnover (timing bug found and fixed)

Benchmarks: SPY; **true equal-weight historical members**.

Monthly equity *and* frozen holdings snapshots saved.

Results: 2008–2022 walk-forward

	Total	CAGR	Vol	Sharpe	Max DD
Tilt (paper)	+196%	7.6%	19.2%	0.48	−35.9%
SPY	+284%	9.5%	16.3%	0.64	−46.3%
Equal-weight	+327%	10.3%	13.3%	0.81	−27.0%

2022: tilt −18.7%, SPY −18.2%, EW −4.2%.

Why the tilt fails — mechanism

- ▶ $\text{corr}(\text{predicted } \Delta \hat{c}_b, \text{realized bin return}) = \mathbf{0.009}$
- ▶ Perfect-foresight variant (realized M): Sharpe 0.47 — no better
- ▶ **Advection mass flows are reclassification, not price pressure:**
crashing stocks are reclassified into the loser bin — it “gains mass” while its members lose money
- ▶ The model predicts *where capital sits*, not *which bins will pay*
- ▶ A return-predictive tilt needs a return model this framework lacks

The paper record does not support real-capital deployment.

- ▶ Distribution model captures conditional spikes ✓
- ▶ Mechanical driver nowcast has genuine skill ✓
- ▶ But the portfolio trails both benchmarks on every risk-adjusted metric ✕
- ▶ **Prerequisite not met.** Protocol, code, and snapshots remain so any future tilt is judged by the same bar.

Next steps

1. **Return-predictive tilt** — auxiliary return model, or crash-risk overlays instead of cross-sectional tilts
2. Within-bin heterogeneity (depth coordinate / particle level)
3. Pre-2009 shares: SEC XBRL starts mid-2009
4. Membership: community reconstruction, day-scale lags possible
5. Damper: assumed parameters need in-sample spike calibration

References

- ▶ fja05680/sp500 — *S&P 500 Historical Components & Changes* (public GitHub reconstruction). Membership.
- ▶ Yahoo Finance — adjusted closes, splits, SPY total return. Prices, benchmark.
- ▶ SEC EDGAR — companyconcept XBRL shares outstanding. Point-in-time shares.

Thank you.